

MINI PROJECT (2024-25)

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By-

Name: Vaibhav Garg

Roll No.: 23UEC638

&

Name: Manikwar Krishna Prabhav

Roll No.: 23UCS640



Department of Electronics and Communication
Engineering

The LNM Institute of Information Technology, Jaipur

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Mini Project

NAME OF PROJECT:

Automated Plant Watering System

AIM:

To design and implement an automated plant watering system that uses sensors and a microcontroller to monitor soil moisture levels and water plants accordingly while also monitoring and displaying the water tank's level using a seven-segment display.

OBJECTIVE:

- To ensure optimal watering of plants by detecting soil moisture levels.
- To automate the process of plant watering to reduce manual effort and water wastage.
- To display the water level in the tank using a proximity sensor and a seven-segment display.
- To study the integration of sensors, actuators, and microcontrollers in real-time applications.

BRIEF INTRODUCTION:

Automated plant watering systems are innovative solutions aimed at minimizing manual intervention in plant care while ensuring adequate irrigation.

The system uses a soil moisture sensor to measure the moisture content of the soil. When the moisture level falls below a predefined threshold, the sensor triggers a water pump to irrigate the soil for a fixed duration. Simultaneously, a proximity sensor monitors the water level in the tank, displaying the level on a seven-segment display for easy monitoring.

This system combines hardware components like sensors, relays, and an Arduino microcontroller with efficient logic to achieve automation.

COMPONENTS USED:

- Soil Moisture Testing Sensor
- ADC Comparator
- Ultrasonic Distance Sensor
- Arduino Microcontroller
- 5V Relay Module
- Seven-Segment Display
- Water Pump
- Power Supply Unit
- Jumper Wires and Breadboard
- Resistors.
- Supporting Frame and Tank

THEORY:

The automated plant watering system incorporates various components that work together to ensure efficient and consistent operation. The soil moisture sensor measures the volumetric water content in the soil and generates an analog signal corresponding to the moisture level. This signal is processed by an ADC comparator, which converts the analog signal into a digital output (0 or 1). When the moisture level drops below the threshold, the comparator outputs a digital LOW signal, triggering the watering process.

The Arduino microcontroller serves as the system's central unit, receiving inputs from the sensors and controlling the outputs to the relay and display. A 5V relay module acts as an interface between the Arduino and the water pump, enabling the microcontroller to switch the pump on and off safely. The relay activates the pump for a predefined duration (e.g., 3 seconds) to deliver water.

For monitoring the water tank, a proximity sensor is used to measure the distance between the sensor and the water surface. Based on this distance, the microcontroller calculates the water level and updates the seven-segment display. The display provides a numeric representation of the water level, with values ranging from 0 (empty) to 5 (full). This ensures real-time feedback for users about the tank's status.

By integrating these components, the system automates plant irrigation and provides an intuitive method to monitor water resources, ensuring convenience, efficiency, and resource conservation.

Understanding the Moisture Sensor

1. Principle of Operation:

The moisture sensor consists of two metal probes inserted into the soil. These probes measure the soil's electrical conductivity, which depends on its moisture content:

- Dry Soil: High resistance, as water content is low.
- Moist Soil: Low resistance, as water increases conductivity.

2. Sensor Output:

The sensor generates an analog voltage signal corresponding to the soil's moisture level:

- Low Voltage: Indicates higher soil moisture.
- High Voltage: Indicates dry soil.

3. Microcontroller Processing:

- The microcontroller reads the analog signal from the moisture sensor.
- The analog signal is converted to a digital value using an analog-to-digital converter (ADC).
- This digital value is compared to a predefined threshold to classify the soil as dry or moist.

4. Triggering the Watering System:

- Dry Soil Detected: If the digital value exceeds the threshold (indicating dryness), the microcontroller activates a relay module.
- The relay switches on a water pump to deliver water to the plant.
- Moist Soil Detected: The pump is turned off when the moisture level exceeds the optimal threshold, preventing overwatering.

This process ensures precise irrigation, conserving water while maintaining optimal soil moisture for plant health.

Understanding the Water Pump

1. Types of Water Pumps:

- DC Water Pump:
 - Ideal for small-scale projects due to its compact size, low voltage operation (commonly 5V, 12V, or 24V), and easy integration with microcontrollers and battery-powered systems.
- AC Water Pump:
 - Suitable for larger or more demanding applications due to higher power output.
 - Requires a voltage converter or relay for safe control when integrated into microcontroller-based systems.

2. Pump Operation:

- The pump is activated via the relay module, which acts as a switch controlled by the microcontroller.
- When the relay is triggered:

- The pump draws water from a reservoir or source.
- It delivers water through tubing to the plant or desired target.
- Once the task is complete or moisture conditions are met, the relay switches the pump off.

3. Flow Rate Control:

- Using a Valve: A valve in the tubing can physically restrict water flow to adjust the rate.
- Adjusting Voltage: In DC pumps, reducing the supplied voltage lowers the pump speed, thereby decreasing the flow rate.

4. Power Consumption:

- Factors Influencing Power Use:
 - Voltage (V): Determines the pump's driving force.
 - Current (I): Depends on the workload and resistance in the system.
 - Operating Time: Directly affects energy usage.
- Efficiency Considerations:
 - Select a pump rated for the specific project demands.
 - Prioritize energy-efficient models to minimize operational costs and environmental impact.

This understanding ensures the selection and operation of the water pump align with project goals, enabling efficient and reliable irrigation.

Understanding the Relay Module

1. What is a Relay Module?

- A relay module is an electronic switch that allows a low-power control signal (from a microcontroller) to operate a high-power electrical device.
- It acts as an interface between two electrically isolated circuits, enabling safe control of high-voltage or high-current devices like water pumps.

2. How it Works:

- Signal from Microcontroller:
The microcontroller sends a low-voltage control signal (e.g., 5V) to the relay module.
- Signal Amplification:
The relay module processes this signal, energizing its internal switch mechanism.

- **Circuit Switching:**
The relay then closes or opens the high-power circuit, controlling devices such as pumps or motors.

3. Why is it Needed?

- **Voltage Handling:**
Microcontrollers are designed for low-voltage (3.3V–5V) operation, while devices like water pumps require higher voltages (e.g., 12V DC or 230V AC).
- **Electrical Isolation:**
The relay ensures the low-voltage control circuit is electrically isolated from the high-voltage load, protecting the microcontroller from potential damage.

4. Types of Relay Modules:

- **Electromagnetic Relay (EMR):**
 - Operates using an electromagnet to mechanically open or close contacts.
 - **Pros:**
 - Cost-effective and widely used.
 - Handles higher current ratings.
 - **Cons:**
 - Slower switching speed.
 - Prone to mechanical wear over time.
- **Solid-State Relay (SSR):**
 - Uses semiconductor components (like thyristors or triacs) to switch circuits electronically.
 - **Pros:**
 - Faster switching speeds.
 - Longer lifespan due to no moving parts.
 - Noise-free operation.
 - **Cons:**
 - Typically more expensive than EMRs.
 - Can generate heat during operation.

By incorporating a relay module into the system, it becomes possible to bridge the gap between low-power control and high-power functionality, ensuring both efficiency and safety in the project.

PIN CONNECTIONS:

1. Relay Board:
 - Signal Pin: Digital Pin 2 (Relay control to manage the water pump)
 - VCC: 5V of Arduino
 - GND: GND of Arduino
2. Soil Moisture Sensor:
 - Signal Pin: Digital Pin 3
 - VCC: 5V of Arduino
 - GND: GND of Arduino
3. Ultrasonic Sensor (for water level monitoring):
 - Trigger Pin: Digital Pin 4
 - Echo Pin: Digital Pin 5
 - VCC: 5V of Arduino
 - GND: GND of Arduino
4. Seven-Segment Display:
 - Segment A: Digital Pin 6
 - Segment B: Digital Pin 7
 - Segment C: Digital Pin 8
 - Segment D: Digital Pin 9
 - Segment E: Digital Pin 10
 - Segment F: Digital Pin 11
 - Segment G: Digital Pin 12
 - Common Cathode: GND of Arduino

WORKING:

1. The soil moisture sensor continuously monitors the moisture content in the soil.
2. When the soil moisture level falls below the threshold, the ADC comparator sends a digital LOW signal to the Arduino.
3. The Arduino processes this signal and activates the relay, powering the water pump for 3 seconds.
4. Simultaneously, the proximity sensor at the top of the water tank detects the water level by measuring the distance to the water surface.
5. Based on the proximity sensor's data, the Arduino determines the tank's water level and updates the seven-segment display with a value between 0 (empty) and 5 (full).
6. The cycle repeats, ensuring consistent soil moisture levels and real-time monitoring of the tank's water level.

FLOW OF EXECUTION:

1. Initialize all components and sensors in the Arduino code.
2. Continuously read data from the soil moisture sensor and proximity sensor.
3. Check if the soil moisture level falls below the threshold.
 - If yes, activate the relay to turn on the water pump for 3 seconds.
4. Measure the water level in the tank using the proximity sensor.
5. Update the seven-segment display with the current water level.
6. Loop the process to ensure real-time operation.

PROJECT CODE:

```
int water;  
int ultraSonic = 0;  
int relay = 2;  
int soilSensor = 3;  
int trigger = 4;  
int echo = 5;  
int a = 6, b = 7, c = 8, d = 9, e = 10, f = 11, g = 12;
```

```
void setup()  
{  
  Serial.begin(9600);  
  pinMode(relay, OUTPUT);  
  pinMode(soilSensor, INPUT);  
  pinMode(trigger, OUTPUT);  
  pinMode(echo, INPUT);  
  pinMode(a, OUTPUT);  
  pinMode(b, OUTPUT);  
  pinMode(c, OUTPUT);  
  pinMode(d, OUTPUT);  
  pinMode(e, OUTPUT);  
  pinMode(f, OUTPUT);  
  pinMode(g, OUTPUT);  
  digitalWrite(relay, LOW);  
  resetDisplay();  
}
```

```
void resetDisplay()  
{  
  // Turn off all segments  
  digitalWrite(a, LOW);  
  digitalWrite(b, LOW);  
  digitalWrite(c, LOW);  
  digitalWrite(d, LOW);  
  digitalWrite(e, LOW);  
  digitalWrite(f, LOW);
```

```
    digitalWrite(g, LOW);  
}
```

```
void displayDigit(int dig)
```

```
{  
    switch (dig)  
    {  
    case 0:  
        digitalWrite(a, HIGH);  
        digitalWrite(b, HIGH);  
        digitalWrite(c, HIGH);  
        digitalWrite(d, HIGH);  
        digitalWrite(e, HIGH);  
        digitalWrite(f, HIGH);  
        break;  
    case 1:  
        digitalWrite(b, HIGH);  
        digitalWrite(c, HIGH);  
        break;  
    case 2:  
        digitalWrite(a, HIGH);  
        digitalWrite(b, HIGH);  
        digitalWrite(d, HIGH);  
        digitalWrite(e, HIGH);  
        digitalWrite(g, HIGH);  
        break;  
    case 3:  
        digitalWrite(a, HIGH);  
        digitalWrite(b, HIGH);  
        digitalWrite(c, HIGH);  
        digitalWrite(d, HIGH);  
        digitalWrite(g, HIGH);  
        break;  
    case 4:  
        digitalWrite(b, HIGH);  
        digitalWrite(c, HIGH);  
        digitalWrite(f, HIGH);  
        digitalWrite(g, HIGH);  
        break;  
    case 5:  
        digitalWrite(a, HIGH);  
        digitalWrite(c, HIGH);  
        digitalWrite(d, HIGH);  
        digitalWrite(f, HIGH);  
        digitalWrite(g, HIGH);  
        break;  
    }  
}
```

```
long readUltrasonicDistance(int trigger, int echo)
```

```

{
    digitalWrite(trigger, LOW);
    delayMicroseconds(2);
    digitalWrite(trigger, HIGH);
    delayMicroseconds(10);
    digitalWrite(trigger, LOW);
    long duration = pulseIn(echo, HIGH, 30000);
    if (duration == 0)
    {
        return -1;
    }
    return (duration * 0.034 / 2);
}

void loop()
{

    water = digitalRead(soilSensor);

    if (water == LOW)
    {
        digitalWrite(relay, HIGH);
    }
    else
    {
        digitalWrite(relay, LOW);
    }

    long distance = readUltrasonicDistance(trigger, echo);
    ultraSonic = (int)distance;
    Serial.print("Distance: ");
    Serial.println(ultraSonic);

    if (ultraSonic > 40)
    {
        displayDigit(0);
    }
    else if (ultraSonic > 30)
    {
        displayDigit(1);
    }
    else if (ultraSonic > 20)
    {
        displayDigit(2);
    }
    else if (ultraSonic > 10)
    {
        displayDigit(3);
    }
    else if (ultraSonic > 5)

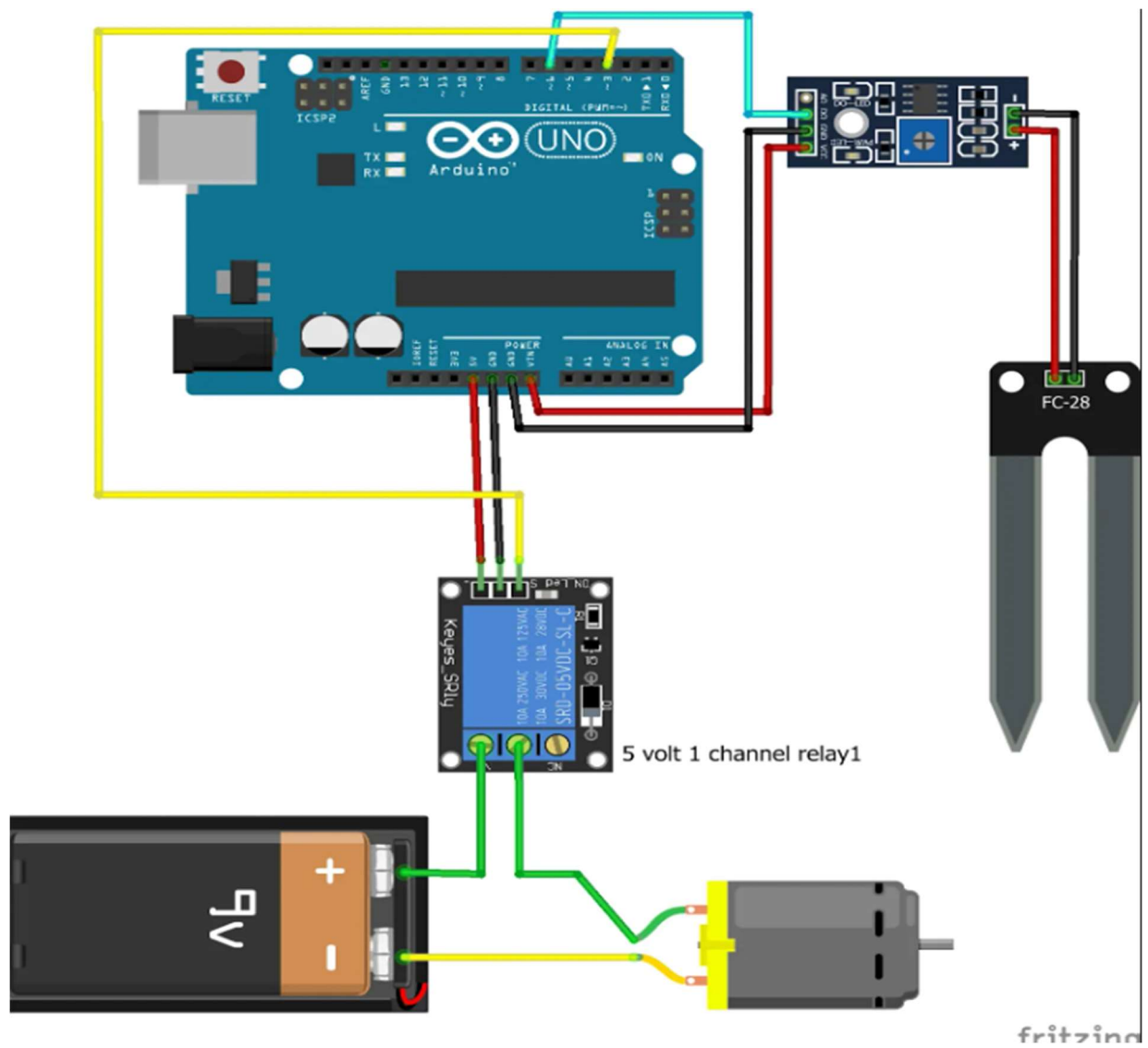
```

```

{
  displayDigit(4);
}
else
{
  displayDigit(5);
}

delay(3000);
}

```



FURTHER ADVANCEMENT:

1. Integration with IoT platforms to monitor and control the system remotely.
2. Use of solar power for energy efficiency.
3. Addition of multiple soil moisture sensors for larger-scale applications.
4. Implementation of a feedback mechanism to adjust watering duration based on weather conditions.
5. Enhanced display features with an LCD or touchscreen interface.

REAL-LIFE APPLICATIONS:

1. Urban gardening and household plant care.
2. Precision agriculture for efficient water usage.
3. Greenhouses and controlled-environment farming.
4. Automated landscaping in commercial or residential areas.
5. Smart irrigation systems for public parks.

CONCLUSION:

The automated plant watering system is an effective and efficient solution for maintaining soil moisture levels and monitoring water resources. By automating the irrigation process, the system reduces manual effort, prevents overwatering, and conserves water. Its modular design and scope for further advancements make it a versatile tool for modern agricultural and gardening practices.